IMAGE PROCESSING TECHNIQUES – EXTENDED CLASS ACTIVITIES WORKSHEET

UG Level | Duration: 10 Minutes per Activity
Name: Aditya Chauhan

Reg No: RA2311003030068

Date: 09/09/2026

Activity 16

he "Sliding Window" Analogy (Manual Harris Corner Detector)

- **Concept:** Corner detection works by sliding a small window over an image and looking for large intensity changes in all directions.
- **Activity:**
    1. Take a printed, high-contrast black-and-white image (e.g., a simple drawing of a house).
    2. Cut a small square hole (e.g., inches) in a piece of black cardboard to act as the "window."
    3. Slide this window across the image.
    4. **Observation:**
        - **Flat Area:** Moving the window shows only white or black, so no intensity change.
        - **Edge:** Moving the window along the edge shows change in one direction.

Corner: Moving the window in any direction shows a change in intensity, signaling a corner.
Observation:

1. Flat Area: Moving the window shows only white or black pixels, so there is little or no intensity change.
2. Edge: Moving the window across an edge shows a significant intensity change mainly in one direction.
3. Corner: Moving the window in any direction shows a significant change in intensity, indicating the presence of a corner.

Conclusion: A corner is detected when the image intensity changes significantly in multiple directions as the window moves over the image.